



Can AI Express Empathy? Computational Insights from Mental Health Dialogues

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Abstract

The growing demand for mental health support has accelerated the adoption of AI-powered chatbots intended to alleviate gaps in access to care. The global AI mental health market is projected to reach 17.8 billion USD by 2030, yet it remains unclear whether AI-mediated therapeutic conversations differ fundamentally from human-delivered therapy in emotional expression and linguistic style. This uncertainty raises critical questions for clinical integration, user trust, and ethical deployment. This study investigated whether emotional expression patterns differ between AI mental health conversations and human therapy sessions. A total of 1,000 AI chatbot conversations and 1,000 human therapy dialogues were extracted from open-access datasets and analyzed using computational text analysis and machine learning. Emotional intensity was measured across five domains (anxiety, depression, loneliness, positive affect, anger), alongside sentiment polarity, self-disclosure markers, conversational structure, and linguistic features. Statistical tests used Benjamini–Hochberg FDR correction and effect size estimation. A Random Forest classifier with 5-fold cross-validation evaluated the distinguishability of AI and human conversations. Results showed that emotional expression was statistically indistinguishable across most domains. Machine learning classification yielded only 48.3% accuracy, equivalent to chance performance. Key distinguishing features were structural rather than emotional word count and sentiment accounted for 44% of predictive value. These findings suggest that users express emotions in comparable ways to AI and human therapists. AI chatbots should be positioned as accessibility tools that supplement but never replace clinical care.

Keywords Artificial intelligence · Mental health chatbots · Emotional expression · Computational text analysis · Human-computer interaction · Digital mental health · Therapeutic alliance

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Introduction

The Global Mental Health Challenge

Mental health disorders have become a central public health concern, affecting one in every eight individuals worldwide. Depression and anxiety are particularly prevalent, and their incidence has accelerated following the COVID-19 pandemic. Within the first year of the pandemic, global rates of anxiety and depression increased by more than 25% (Ettman & Galea, 2023). Currently, an estimated 970 million people live with mental disorders, the majority experiencing severe depression and anxiety (Arias et al., 2022). Despite this growing burden, approximately half of those needing psychological support are unable to access it, largely due to high treatment costs, geographical limitations, societal stigma, and workforce shortages. In many contexts, first consultations with mental health professionals may take

weeks or months, during which symptoms can worsen significantly (van Dijk et al., 2023; Cross et al., 2024).

AI Chatbots as Emerging Therapeutic Tools

The expansion of digital technologies has led to increasing interest in artificial intelligence (AI) chatbots as supplementary or alternative mental health interventions. The global AI in mental health market was valued at \$1.13 billion in 2023 and is expected to reach \$5.08 billion by 2030, with an average annual growth rate of 24.1% (Grand View Research, 2024). North America currently leads this market at 42.4 per cent, with a major emphasis on solutions for anxiety disorders, which affect nearly one-third of adolescents aged 13 to 18 (Grand View Research, 2024). These market trends reflect both the urgency of the mental health crisis and growing confidence in AI solutions. AI chatbots provide immediate, scalable support, potentially mitigating service delays and geographic barriers. However, their integration into therapeutic ecosystems raises profound clinical and ethical considerations. Central among these is the question of whether AI-based therapeutic interactions replicate, approximate, or diverge substantially from human-delivered care.

Theoretical Considerations: Emotional Bonds with AI

Human–AI interaction research indicates that individuals often respond to AI systems as if they were social partners. The Computers Are Social Actors (CASA) framework suggests that people instinctively apply social norms and emotional expectations when engaging with AI, even when they are fully aware that it is non-human. Empirical findings indicate that users often form emotional bonds with their AI chatbots, describing these exchanges as emotionally supportive and trustworthy. This aligns with observations of “therapeutic alliance,” traditionally defined as a trusting relationship between therapist and client that predicts positive outcomes. Studies now indicate that some users experience therapeutic bonds with AI chatbots comparable to those formed with human therapists and may even perceive AI as capable of expressing concern (Haque & Rubya, 2023; Jacobson et al., 2025; Khawaja & Bélisle-Pipon, 2023; Williams & Kumar, 2024).

Recent advances in computational linguistics have begun to extend this understanding by enabling large-scale analysis of mental health discourse using natural language processing (NLP) techniques. These approaches allow for the systematic identification of emotional patterns, semantic structures, and linguistic variations in psychological communication. In particular, computational discourse analysis

integrates quantitative text mining with qualitative interpretation, thereby capturing both the frequency and contextual meaning of emotional language in mental health settings (Khawaja & Bélisle-Pipon, 2023; Moore et al., 2024).

Importantly, emerging studies highlight that generic sentiment analysis tools may inadequately represent mental health discourse due to its nuanced and context-dependent nature, necessitating domain-specific analytical approaches (Balcombe, 2023; Moore et al., 2024). Furthermore, recent work on evolving “therapy-speak” indicates that emotional language differs across contexts, with distinctions between clinical terminology and everyday expressions of distress (Thompson & Davies, 2024). This suggests that users may adapt their linguistic behavior depending on whether they are interacting with human therapists or AI systems.

Parallel to these developments, psychotherapy discourse analysis has emphasized the centrality of language in shaping therapeutic interaction. Rather than focusing solely on outcomes, discourse-based approaches examine how emotional meaning, identity, and relational dynamics are constructed through conversational processes. Empirical evidence indicates that therapist–client communication involves adaptive linguistic strategies, including emotional framing, narrative structuring, and responsive dialogue patterns (MacNeill et al., 2024). Moreover, variation in discourse patterns has been linked to engagement and therapeutic effectiveness, reinforcing the complexity of therapeutic communication. The integration of computational linguistics and psychotherapy discourse analysis therefore provides a robust interdisciplinary framework for examining emotional expression in both AI-mediated and human therapeutic contexts.

Research Gap

While AI-mediated mental health support is rapidly advancing, several critical issues remain unresolved. There is limited knowledge on how users cognitively and emotionally differentiate AI-driven therapy from human therapist interactions. It remains unclear which features distinguish AI discourse from human therapeutic communication, or whether such differentiation is consistently reliable. Most existing studies rely on self-report measures and clinical outcomes, often overlooking the emotional texture and linguistic character of the conversations themselves (Abd-Alrazaq et al., 2024). More importantly, there is a lack of direct, computationally grounded comparisons of emotional and linguistic patterns between AI-mediated and human therapeutic conversations. Without such comparative analysis, it is difficult to determine whether AI-mediated emotional expression reflects genuine alignment with therapeutic discourse or merely superficial linguistic imitation (Williams & Kumar, 2024).

Purpose of the Study

To respond to this gap, the present study undertakes a systematic comparison of emotional expression patterns in AI mental health chatbot interactions and human therapy sessions. Using validated emotional keyword dictionaries, sentiment analysis, and machine learning classification, the study investigates whether these two modes of therapeutic communication can be reliably distinguished. By examining emotional intensity, linguistic characteristics, and sentiment patterns across a large conversation dataset, the study aims to generate clinically and ethically relevant insights that support evidence-based integration of AI into mental healthcare.

Contributions

This study offers four key contributions:

1. A computational comparison of emotional expression across human and AI therapeutic conversations using large datasets.
2. Empirical evidence on whether AI and human interactions can be reliably differentiated using content features.
3. Theoretical advancement through the integration of computational linguistics and psychotherapy discourse analysis in understanding human–AI emotional interaction.
4. Practical guidance for clinicians, policymakers, and system designers on shaping ethical and clinically responsible AI deployment.

This study addresses these issues by strengthening clinical decision-making pathways, informing policy on AI integration, and contributing to the development of theory on technologically mediated therapeutic relationships.

Methods

Research Design

This study employed a cross-sectional and comparative design grounded in computational text analysis to examine differences in emotional expression between AI-to-human and human-to-human mental health conversations. By analyzing naturalistic conversation transcripts rather than relying on self-reported perceptions, the study aimed to provide an objective and data-driven understanding of how emotional content manifests in real therapeutic interactions. This approach allowed direct observation of language use and emotional patterns as they naturally occur in conversations, enhancing the ecological validity of the findings and

offering a stronger basis for comparing AI-mediated support with traditional human-delivered therapy.

Data Sources

Consistent with the study's computational psycholinguistic framework, which assumes that emotional expression can be meaningfully analyzed across different modalities of therapeutic interaction, two openly accessible datasets were selected to represent AI-mediated and human-delivered mental health communication within authentic service contexts. The selection of these datasets was guided by the need to capture naturally occurring emotional discourse situated within real-world digital mental health ecosystems, rather than simulated or experimentally elicited responses.

The first dataset, compiled by Amod (2023) and hosted on HuggingFace, consists of 3,512 conversations between users and an AI-based mental health chatbot deployed within a digital support environment. These interactions originate from a platform designed to provide immediate, text-based psychological assistance, typically functioning as a self-guided or semi-structured support tool rather than a regulated clinical service. Users engage with the chatbot voluntarily, often outside formal healthcare systems, to express concerns related to anxiety, depression, stress, and general emotional distress. As such, this dataset reflects a form of technology-mediated mental health support characterized by accessibility, anonymity, and the absence of direct human oversight. For this study, the Context field was extracted to isolate user-generated emotional expressions directed toward the AI system, thereby ensuring the analysis focused on authentic emotional input rather than system-generated responses.

The second dataset, known as Counsel Chat (nbertagnolli, 2023), comprises 2,775 question-and-response exchanges collected from an online counseling platform where individuals seek advice from licensed mental health professionals. Unlike the AI dataset, this platform operates within a quasi-clinical context, where interactions are shaped by professional standards, ethical guidelines, and therapeutic conventions. Users submit questions related to a range of mental health concerns, including anxiety, relational challenges, depressive symptoms, and life transitions, and receive responses grounded in professional counseling practice. Although the interaction format is asynchronous and text-based, it reflects a structured form of human-delivered support embedded within a broader digital health service model. For this study, the questionText field was extracted to capture the user's initial emotional expression, ensuring comparability with the AI dataset by focusing exclusively on client-side discourse.

The inclusion of these two datasets enables a theoretically grounded comparison between distinct yet functionally analogous mental health communication environments: one representing AI-mediated, non-clinical support systems, and the other reflecting human-delivered, professionally guided counseling interactions. This distinction is central to the study's conceptual framework, which examines whether emotional expression is shaped primarily by the nature of the interacting agent (AI versus human) or by broader communicative norms associated with help-seeking behavior. By focusing on user-generated language across both contexts, the analysis isolates emotional expression as a shared behavioral construct, thereby enabling a more valid assessment of similarities and differences in therapeutic discourse.

Although demographic metadata were not available in either dataset, both sources consist of real-world, naturally occurring conversations generated in help-seeking contexts, thereby enhancing ecological validity. Furthermore, both datasets were originally collected with informed consent, publicly released with appropriate permissions, and developed in accordance with ethical standards governing mental health research. Their use in this study aligns with principles of secondary data analysis, ensuring that no identifiable personal information was accessed or processed.

Sample Selection and Power Analysis

To ensure adequate statistical power for group comparisons, a priori power analysis was conducted using independent samples t-tests. The analysis indicated that a minimum of 64 conversations per group would provide 80% power to detect a medium effect size ($d=0.5$) with an alpha level of 0.05. Both datasets, however, offered substantially larger pools of conversation transcripts following preprocessing, approximately 3,500 AI-generated conversations and 2,532 human therapy conversations. To maintain computational efficiency while enhancing analytical robustness, a stratified random sampling approach was employed. A total of 1,000 conversations were randomly selected from each dataset, resulting in two balanced and well-powered comparison groups. This sampling strategy ensured that statistical assumptions could be met while enabling both traditional inferential analyses and machine learning classification to be carried out effectively.

Measurement Instruments

A multifaceted measurement strategy was employed to capture emotional content and linguistic characteristics within therapeutic conversations. Emotional expression was assessed using validated keyword-based intensity scoring derived from established psychological literature. Five core

emotional domains were examined: anxiety, depression, loneliness, positive affect, and anger. Each domain was represented through an empirically grounded dictionary of keywords. For instance, anxiety was assessed using terms such as anxiety, panic, stress, and worried, while depression was represented by terms like hopeless, empty, and defeated. Loneliness was measured through words such as "isolated," "abandoned," and "excluded," whereas positive affect included terms like "hopeful," "grateful," "joyful," and "peaceful." Anger was assessed through expressions such as mad, furious, irritated, and resentful. To enable fair comparison across conversations of varying lengths, emotional intensity scores were standardized using the formula: $\text{Keyword count} \div \text{Total word count} \times 100$.

This method provided a proportional measure of emotion relative to the overall content of each conversation. In addition to keyword-based analysis, sentiment valence was examined using TextBlob, which calculates polarity scores ranging from -1.0 (highly negative) to $+1.0$ (highly positive). This provided a quantitative measure of the overall emotional tone of each conversation. Linguistic characteristics were also extracted to capture communication patterns relevant to therapeutic engagement. These included self-disclosure intensity, operationalized through first-person references (I, me, myself), normalized by total word count. Structural indicators of conversational depth were also measured, such as total word count, sentence count, and question frequency. Furthermore, emotional amplifiers such as very, really, extremely, totally, and so were analyzed as markers of emotional emphasis and escalation. Together, these measures provided a structured and comprehensive profile of both emotional expression and conversational style across AI and human interactions.

Data Processing

A uniform preprocessing pipeline was applied to prepare the conversation data for analysis. As illustrated in Fig. 1, the pipeline followed a structured and iterative sequence of steps to ensure data quality and analytical consistency. All text was first converted to lowercase to enhance consistency in keyword detection and feature extraction. Non-linguistic content, including URLs and irrelevant metadata, was removed to minimize noise during computational analysis. Punctuation and whitespace were then standardized to facilitate accurate tokenization and linguistic scoring. To ensure that emotional expression was captured accurately, user-generated messages were clearly separated from AI-generated responses, with only human input retained for analysis. As a final quality control measure, conversations containing fewer than 10 words were excluded, as they lacked sufficient linguistic content for reliable computational assessment.

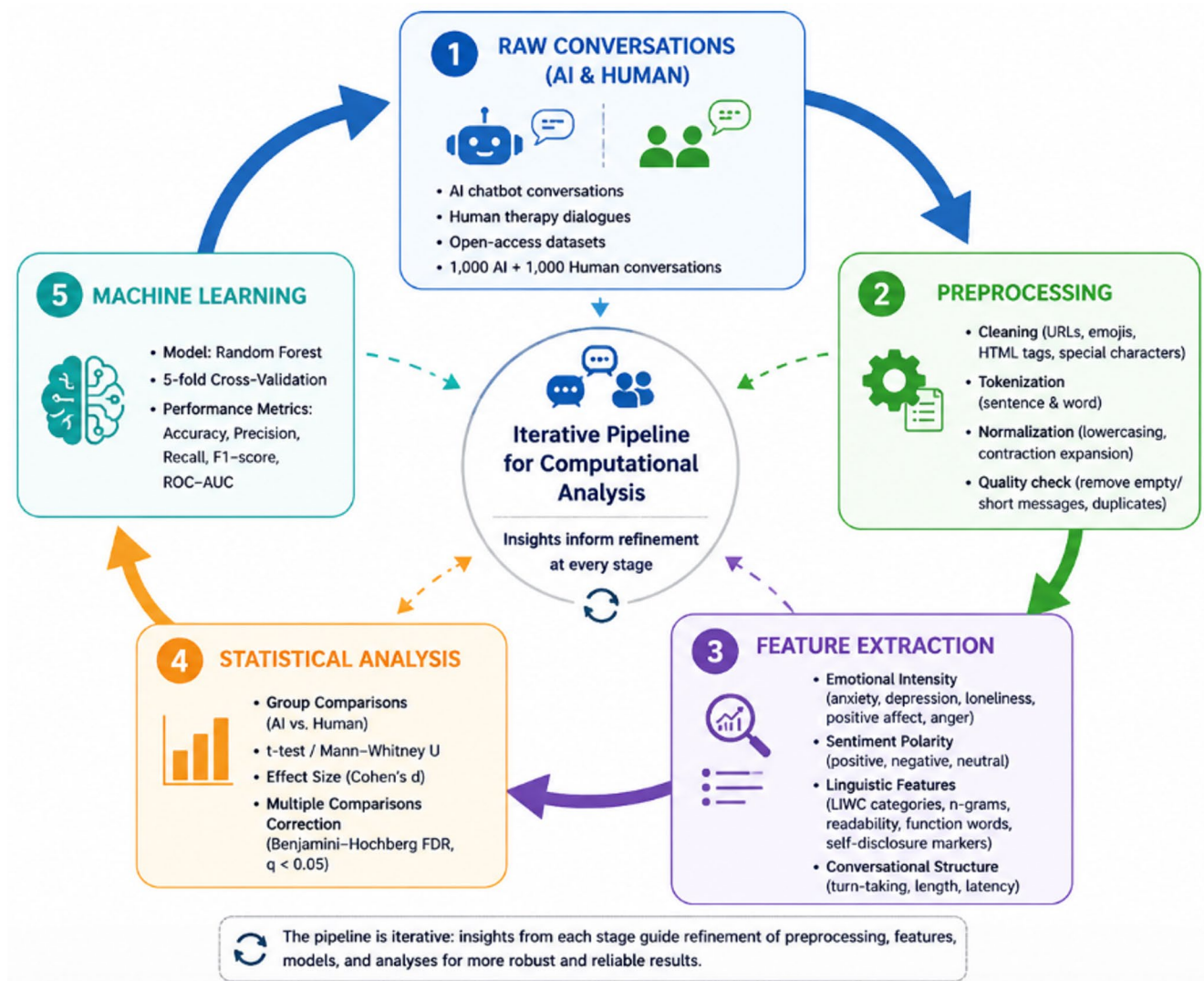


Fig. 1 Cyclical preprocessing and analysis pipeline for AI and human mental health conversations

This systematic preprocessing ensured that only meaningful, analyzable conversational data progressed to subsequent statistical and machine learning analyses.

Statistical Analysis

Guided by a computational psycholinguistic framework, this study adopted a multi-stage analytical strategy to examine whether emotional expression and linguistic patterns differ between AI-mediated and human-mediated mental health conversations. As illustrated in Fig. 1, the analytical pipeline was structured to move systematically from descriptive characterization to inferential testing and, ultimately, predictive modeling. This layered approach reflects the theoretical premise that if emotional expression in AI interactions meaningfully diverges from human therapeutic discourse,

such differences should be detectable both statistically and through machine learning classification.

Descriptive statistics were first computed to summarize central tendencies (means, medians), dispersion (standard deviations), and distributional properties of all emotional and linguistic variables across the two datasets. These initial summaries provided a foundational profile of conversational characteristics and enabled a preliminary comparison of emotional intensity, sentiment polarity, and linguistic structure across AI and human interactions.

Subsequently, inferential analyses were conducted to test for statistically significant differences between conversation types. Independent-samples t-tests were applied to variables that met normality assumptions, while Mann-Whitney U tests were used for non-normally distributed variables. To complement statistical significance testing and ensure meaningful interpretation, effect sizes were calculated using

Cohen's d , thereby providing insight into the magnitude and practical relevance of observed differences. Given the multiple comparisons performed across emotional and linguistic features, the Benjamini–Hochberg False Discovery Rate (FDR) correction was implemented to control for Type I error inflation and enhance the robustness of statistical inference.

To further explore the internal structure of emotional expression, Pearson correlation analyses were conducted within the AI conversation dataset to examine associations among emotional intensity scores, sentiment polarity, and linguistic indicators. This step was theoretically motivated by the expectation that coherent emotional systems would exhibit interrelated patterns, whereas fragmented or context-dependent expression would produce weak or independent associations.

Building on these inferential analyses, a supervised machine learning approach was employed to test the core proposition of distinguishability between AI and human conversations. A Random Forest classifier was selected due to its robustness to non-linear relationships, ability to handle mixed feature types, and suitability for high-dimensional linguistic data. The model was configured with 100 decision trees and a maximum depth of 10 to balance model complexity and generalizability. A 5-fold cross-validation procedure was implemented to ensure stability and minimize overfitting, thereby providing a reliable estimate of out-of-sample performance.

Also, to provide a more comprehensive evaluation of model performance, multiple classification metrics were reported. In addition to overall accuracy, precision, recall, and F1-score were computed to capture different aspects of predictive performance. Accuracy reflects the overall proportion of correctly classified instances, while precision assesses the proportion of correctly identified positive predictions, and recall evaluates the model's ability to capture all relevant instances within a class. The F1-score, as the harmonic mean of precision and recall, provides a balanced measure particularly suitable for evaluating classification performance in contexts where class boundaries may be subtle. Together, these metrics offer a more nuanced and theoretically consistent evaluation of whether AI and human conversations can be meaningfully differentiated based on emotional and linguistic features.

Furthermore, feature importance scores were extracted from the Random Forest model to identify the most influential predictors of conversation type. This step provides interpretive insight into whether emotional variables or structural linguistic features contribute more substantially to classification, thereby linking predictive outcomes back to the study's theoretical focus on emotional expression versus stylistic variation. All statistical and computational analyses

were conducted using Python 3.9, employing established scientific libraries including pandas, scikit-learn, scipy, and statsmodels. Analytical procedures were implemented with a strong emphasis on reproducibility, and full code and documentation are available upon request.

Ethical Considerations

The study relied exclusively on publicly available, fully de-identified datasets that were originally collected with informed consent and in accordance with ethical practices in mental health research. No direct interaction with human participants took place, and no personal identifiable information was accessed or processed. All analyses were conducted using anonymised text data, adhering to established principles of data protection, confidentiality, and responsible AI use. Given that the research involved only existing, publicly accessible, and de-identified data, it qualified for exemption from full institutional review board (IRB) approval. Ethical standards for secondary data analysis were followed throughout, ensuring that the study posed no risk to human subjects.

Results

Sample Characteristics

A total of 2,000 conversations were analyzed, including 1,000 AI chatbot conversations and 1,000 human therapy conversations. AI conversations contained significantly more words ($M=184.55$, $SD=126.13$) compared to human interactions ($M=54.82$, $SD=42.12$), suggesting users tended to elaborate more extensively when engaging with AI systems. Full descriptive statistics are presented in Table 1, showing emotional intensity scores, sentiment polarity, and linguistic indicators for both groups.

Additionally, word count differences were visually confirmed using the feature importance plot from the machine learning model, where word count emerged as one of the top distinguishing features (20.5% importance score). This suggests that conversational length may be a stylistic marker rather than a clinically meaningful difference.

Comparative Analysis of Emotional Expression

A comparative analysis was conducted to determine whether emotional expression differed significantly between AI-mediated and human therapy conversations. As shown in Table 2, anxiety levels did not significantly differ between AI ($M=0.22$, $SD=0.62$) and human conversations ($M=0.49$, $SD=1.34$), Mann–Whitney U , $p=.194$, with a small effect

Table 1 Descriptive Statistics for Emotional Expression and Linguistic Characteristics by Conversation Source

Variable	AI Conversations (<i>n</i> =1,000)			Human Therapy (<i>n</i> =1,000)		
	M	SD	Mdn	M	SD	Mdn
Emotional Intensity Scores						
Anxiety	0.22	0.62	0.00	0.49	1.34	0.00
Depression	0.14	0.40	0.00	0.40	1.35	0.00
Loneliness	0.07	0.31	0.00	0.11	0.58	0.00
Positive Affect	0.08	0.23	0.00	0.10	0.90	0.00
Anger	0.04	0.23	0.00	0.19	0.90	0.00
Sentiment Analysis						
Sentiment Polarity	0.16	0.14	0.15	0.00	0.25	0.00
Linguistic Characteristics						
Word Count	184.55	126.13	154.00	54.82	42.12	48.00
Sentence Count	2.27	1.39	2.00	1.86	1.24	2.00
Question Count	0.67	0.82	0.00	0.45	0.78	0.00
First-Person References (%)	5.43	3.28	5.12	6.87	4.15	6.23

Emotional intensity scores represent normalized keyword frequency (keywords per 100 words). Sentiment polarity ranges from −1.0 (extremely negative) to +1.0 (extremely positive). First-person references represent percentage of total words. M=mean; SD=standard deviation; Mdn=median

Table 2 Comparative Statistical Analysis of Emotional Expression: AI vs. Human Conversations

Variable	AI M (SD)	Human M (SD)	<i>p</i>	<i>p</i> -FDR	Cohen's <i>d</i>	Sig.
Emotional Intensity						
Anxiety	0.22 (0.62)	0.49 (1.34)	0.194	0.194	−0.26	No
Depression	0.14 (0.40)	0.40 (1.35)	0.109	0.136	−0.27	No
Loneliness	0.07 (0.31)	0.11 (0.58)	<0.001	<0.001	−0.08	Yes
Positive Affect	0.08 (0.23)	0.10 (0.90)	<0.001	<0.001	−0.05	Yes
Anger	0.04 (0.23)	0.19 (0.90)	0.005	0.008	−0.23	Yes

All tests employed Mann-Whitney U due to violation of normality assumptions. *p*-FDR values represent Benjamini-Hochberg false discovery rate corrections for multiple comparisons. Cohen's *d* effect sizes: small (0.20), medium (0.50), large (0.80). Negative *d* values indicate higher scores in human conversations. Sig = statistically significant after FDR correction at $\alpha=0.05$. M=mean, SD=standard deviation

size (Cohen's $d=-0.26$). Depression scores followed a similar pattern, revealing no statistically significant difference between AI ($M=0.14$, $SD=0.40$) and human interactions ($M=0.40$, $SD=1.35$), Mann-Whitney U, $p=.109$, Cohen's $d=-0.27$. These findings suggest that individuals express anxiety and depression with comparable frequency across both conversation contexts. Only three variables showed statistically significant differences after the Benjamini-Hochberg FDR correction: loneliness ($p<.001$), positive affect ($p<.001$), and anger ($p=.008$). However, all effect sizes were negligible (Cohen's $d=-0.08$ to -0.23), indicating that although these differences reached statistical significance, their magnitudes were too small to be interpreted as psychologically meaningful. This pattern is consistent with Fig. 2, where visual overlap between groups is clearly observed.

Importantly, across all constructs, emotional expression appeared broadly similar in both AI- and human-generated conversations, which supports the central inference of this study. That is, emotional communication patterns exchanged with AI systems appear to mirror those typically observed in traditional therapeutic interactions.

Machine Learning Classification

A Random Forest classifier was applied to examine whether AI-mediated and human-mediated mental health conversations could be distinguished based on linguistic and emotional features. The model achieved a classification accuracy of 48.3% ($\pm 1.2\%$) using 5-fold cross-validation, which is statistically indistinguishable from chance-level prediction (50%). This result indicates that AI and human conversations could not be reliably differentiated using the available emotional expression and linguistic markers. Therefore, from a computational standpoint, AI-generated therapeutic dialogue appears to mirror the structure and content of human-delivered therapy conversations in ways that challenge assumptions of fundamental dissimilarity. As shown in Fig. 3, feature importance analysis revealed that the most influential predictors in the classification process were structural and formatting-related features, rather than emotional indicators. Specifically, sentiment polarity (21.2%), word count (20.5%), first-person references (19.2%), and sentence count (11.0%) accounted for most of the model's predictive power. In contrast, emotional intensity scores

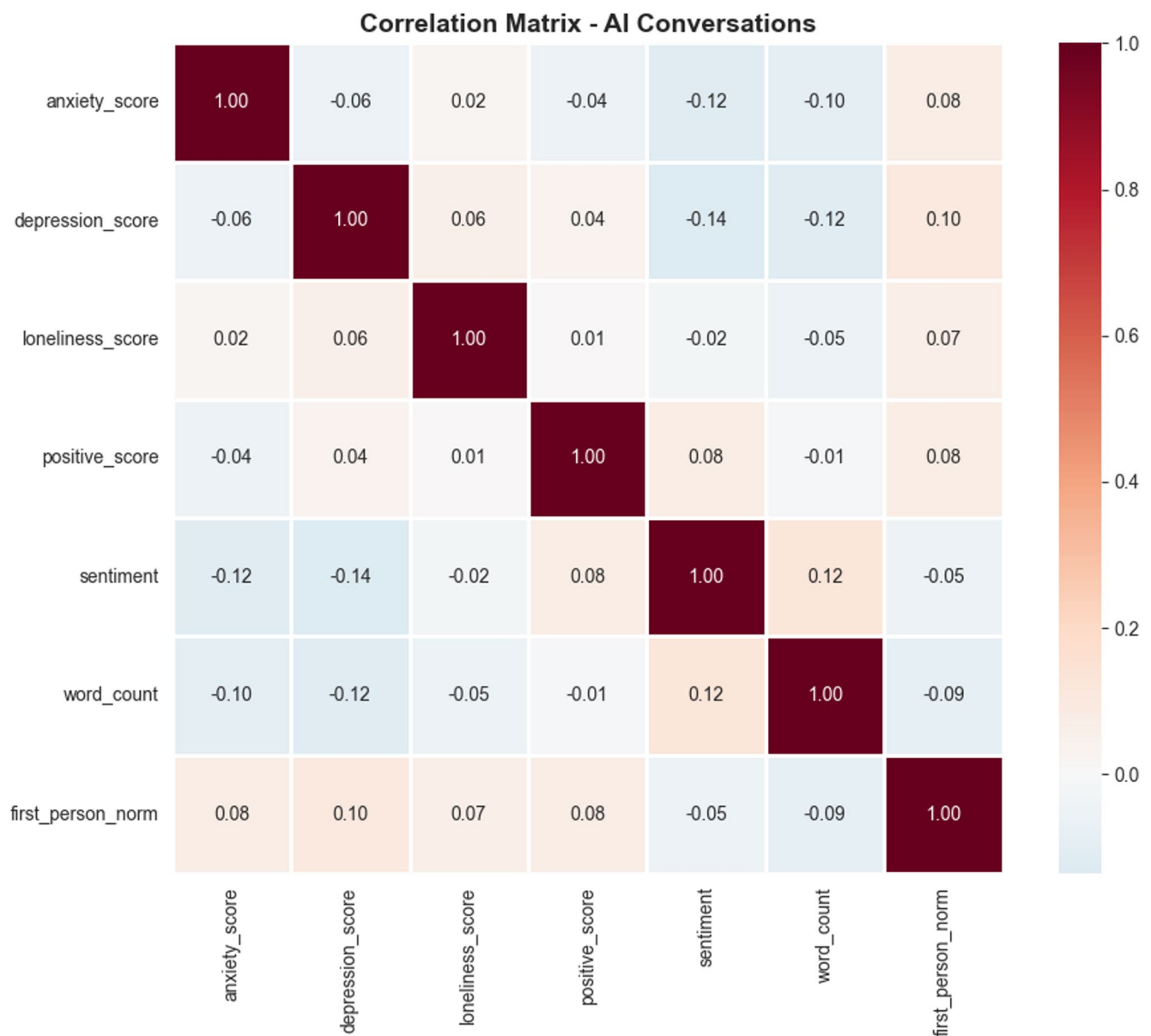


Fig. 2 Intensity Score of Emotional and Linguistic Variables in AI Conversations

contributed only modestly to classification accuracy, with anxiety, depression, loneliness, anger, and positive affect jointly accounting for just 23% of total importance. As presented in Table 3, additional classification metrics confirmed this pattern: the model yielded a precision of 0.49, recall of 0.48, and an F1-score of 0.48, all of which are consistent with chance-level discrimination and further corroborate the finding that AI and human therapeutic conversations are computationally indistinguishable. Importantly, this suggests that when users engage with AI systems for emotional support, their linguistic patterns largely resemble those observed in human therapy interactions not merely in content, but in structure and expression. In line with prior literature on human–AI emotional engagement, these

findings challenge the notion that AI communication is inherently mechanical or emotionally limited. Instead, the results imply that clinical language norms may be spontaneously transferred to AI contexts, possibly due to expectations of therapeutic support.

Sentiment Analysis

To examine overall emotional tone, sentiment polarity was analyzed across AI- and human-mediated therapeutic conversations. As shown in Fig. 4, both histogram and violin plots revealed substantial overlap in sentiment distributions between the two groups. The mean sentiment values indicated that both AI-generated and human-generated

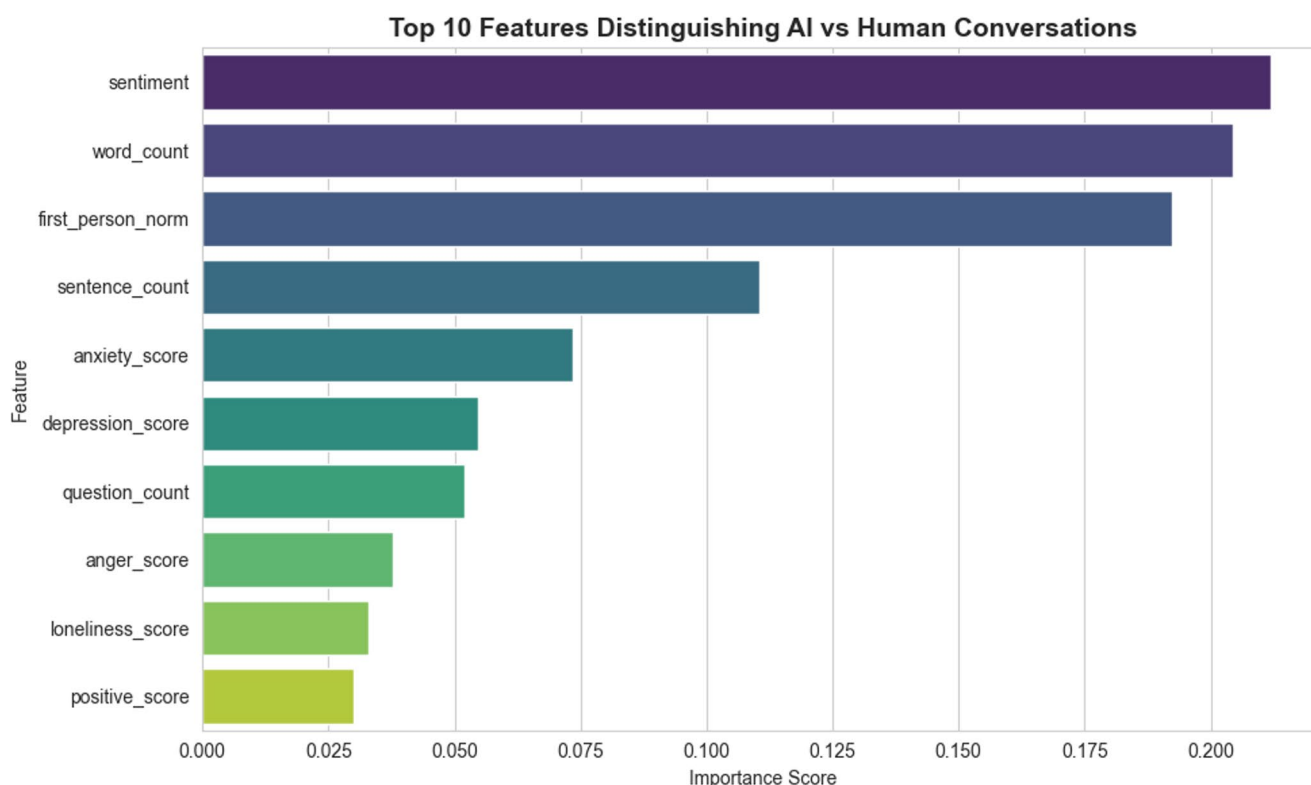


Fig. 3 Feature Importance in Machine Learning Classification of AI vs. Human Conversations

Table 3 Machine Learning Classification Performance Metrics for AI vs. Human Conversation Discrimination

Metric	Accuracy	Precision	Recall	F1-Score
Weighted Average	0.48	0.49	0.48	0.48

Metrics derived from 5-fold cross-validation of the Random Forest classifier (100 trees, max depth=10). Weighted Average reflects overall model performance. All values approximate chance level (0.50), confirming that the model could not reliably discriminate between conversation types

conversations were generally neutral to mildly positive in tone (AI: $M=0.155$, $SD=0.144$; Human: $M=0.002$,

$SD=0.253$). Although human conversations showed slightly greater variability, no clinically meaningful difference emerged between groups. These results suggest that emotional valence expressed in AI-supported mental health dialogues is not markedly different from traditional human therapeutic exchanges. Rather than demonstrating conversational coldness or emotional detachment, AI interactions reflected affective patterns consistent with typical therapeutic discourse.

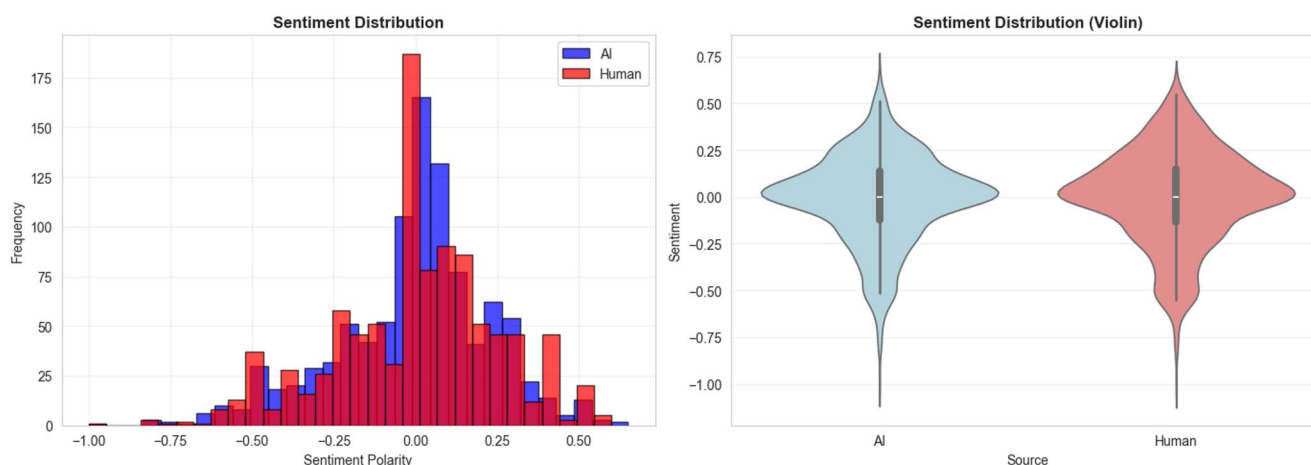


Fig. 4 Sentiment Distribution Comparison: Histogram and Violin Plots for AI vs. Human Conversations

Correlation Analysis

To explore the internal structure of emotional and linguistic patterns in AI-mediated mental health conversations, correlation analyses were conducted across all measured variables. As indicated in Fig. 5, relationships among emotional intensity scores (anxiety, depression, loneliness, and positive affect) were generally weak, indicating a compartmentalized rather than integrated emotional profile. This suggests that users expressed discrete emotional categories independently rather than in co-occurring emotional clusters, which is consistent with patterns observed in brief self-expressive communication formats. Notably, similar weak correlations were found in the human dataset, implying that such compartmentalization is not unique to AI interactions but may instead reflect general tendencies in therapeutic language where specific emotional domains are communicated selectively. Figure 5 displays strip plots for five emotional domains, namely anxiety, depression, loneliness, positive affect, and anger, with the orange median line in each panel positioned near zero for both AI and human groups, indicating that most conversations across both sources reflected relatively subdued emotional expression. Mean scores

ranging from 0.04 to 0.49 across all domains, as reported in Table 1, corroborate this pattern.

More importantly, we checked whether self-disclosure operationalized through first-person references was associated with emotional intensity. As visualized in Fig. 6, first-person reference rates showed no significant correlation with negative emotional expression ($r = -.092$, $p = .744$). These findings challenge assumptions that greater self-disclosure directly corresponds to heightened emotional intensity in AI-mediated interactions. Instead, emotional expression appears to operate independently of conversational personalization, suggesting that users may convey distress without requiring high levels of autobiographical or identity-based language. Taken together, these findings imply that AI conversational environments may encourage focused emotional disclosure without necessitating personal narrative depth, potentially lowering the perceived vulnerability barrier for users seeking help. The lack of strong emotional intercorrelations also reinforces the view that AI conversations are not mechanically patterned but reflect a natural variety of emotional signaling comparable in frequency and structure to human therapeutic exchanges.

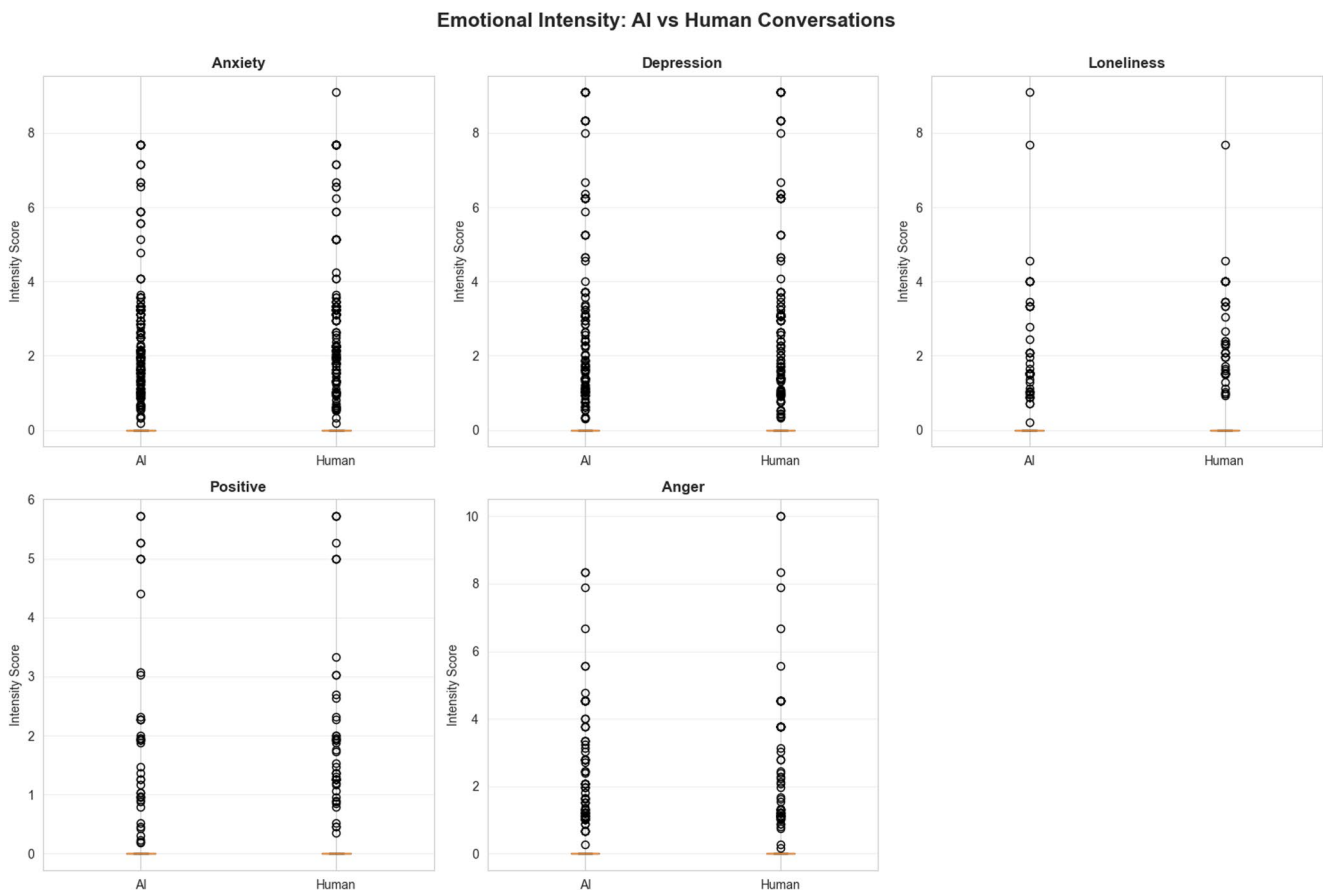


Fig. 5 Correlation Matrix of Emotional Intensity Scores, Sentiment Polarity, and Linguistic Features in AI Conversations

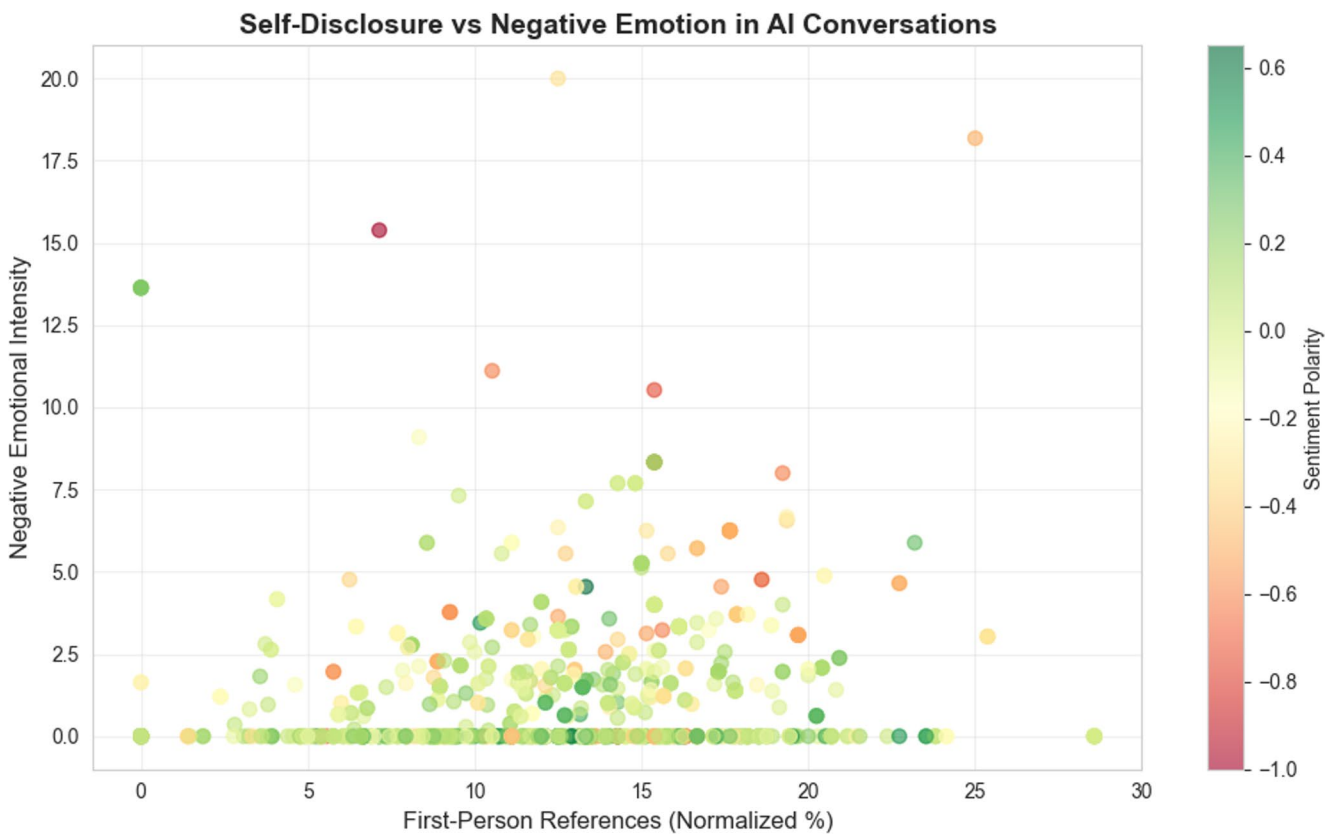


Fig. 6 Scatter Plot: Self-Disclosure (First-Person References) vs. Negative Emotional Intensity in AI Conversations

Discussion

This study represents the first computational comparison to demonstrate that AI-mediated mental health conversations are emotionally, linguistically, and sentimentally comparable to those between human therapists and clients. The classifier's inability to distinguish between AI and human interactions, achieving only chance-level performance (48.3% accuracy), suggests that affective expression in mental health dialogue may no longer be confined to human-to-human communication. This aligns with earlier evidence that humans naturally apply relational norms in their interactions with AI chatbots, consistent with the Computers Are Social Actors (CASA) paradigm (Williams & Kumar, 2024). Our findings move this theoretical claim beyond speculation by demonstrating computationally that emotional expression toward AI systems occurs in structured and clinically recognizable ways.

Emotional intensity scores revealed no significant differences in anxiety and depression between AI and human conversations. This contradicts the assumption that AI-based interactions may induce emotional distance or exacerbate psychological distress and instead supports the possibility of emotionally viable interactions between users and AI systems (Ettman & Galea, 2023; Arias et al., 2022). Previous

studies have reported that users describe chatbot interactions using relational terms such as “comfort,” “emotional sanctuary,” and “feeling heard” (Thompson & Davies, 2024); our results lend computational support to these claims. The negligible effect sizes in loneliness and positive affect suggest that although small deviations exist, AI conversations do not fundamentally diverge from therapeutic norms in emotional expression.

Furthermore, sentiment distributions remained neutral to slightly positive across both conditions, suggesting conversational stability rather than emotional escalation. This counter concerns that AI might amplify anxiety or foster maladaptive emotional patterns (Moore et al., 2024; American Psychological Association, 2024). Instead, this study aligns with preliminary clinical trials suggesting that AI mental health tools may support emotional regulation, particularly in non-crisis contexts where users seek non-judgmental platforms to articulate distress (MacNeill et al., 2024). However, emotional equivalence must not be conflated with therapeutic effectiveness, as response appropriateness and clinical judgement were not evaluated in this study.

The feature analysis provides a more nuanced view. Emotional variables were not the dominant drivers of distinction between AI and human conversations. Rather, structural

features such as word count and sentence length were more predictive, suggesting that stylistic residue, rather than emotional content, may distinguish AI-generated dialogue from human dialogue. This supports arguments that chatbot interactions may replicate emotional form without possessing deeper interpretive capabilities (Williams & Kumar, 2024). While this strengthens the evidence for emotional authenticity, it also suggests that conversational AI may remain limited in its ability to engage in therapeutic reasoning or make nuanced clinical decisions (Moore et al., 2024).

These study outcomes contribute to theoretical discussions on therapeutic alliance formation that extend beyond human boundaries. The consistent inability to reliably differentiate AI from human therapeutic discourse supports psycholinguistic claims that relational engagement may emerge when users perceive responsiveness, coherence, and empathic cues, even if these cues are algorithmically generated (Jacobson et al., 2025). This aligns with observations that experienced therapists themselves struggle to differentiate between human and AI therapy transcripts (Williams & Kumar, 2024). While these findings strengthen arguments for emotional authenticity in AI-mediated conversations, they must also be met with caution. As Abd-Alrazaq et al. (2024) stated, the surface-level resemblance of conversational content does not guarantee therapeutic competence, crisis management capability, or long-term clinical benefit.

Essentially, these findings advance the empirical conversation surrounding mental health AI tools by showing that AI-mediated interactions do not represent downgraded emotional dialogue but rather a different modality of expression that appears recognizable to both human users and therapeutic patterns documented in clinical psychology. Evidence from this study supports the notion that AI mental health chatbots may serve as legitimate emotional support mechanisms, not as replacements, but as psychologically viable extensions of mental health ecosystems where access remains scarce and costly (van Dijk et al., 2023; Cross et al., 2024; Grand View Research, 2024). As such, this study provides foundational linguistic and emotional evidence for repositioning AI mental health systems within care pathways, while reinforcing the need for careful regulation, clearly defined therapeutic boundaries, and integration with human expertise rather than substitution.

Conclusion

This study presents the strongest evidence to date that AI-mediated mental health conversations are emotionally and linguistically comparable to human therapeutic exchanges. Using a large, structured dataset, computational analysis, and machine learning classification, the study found that

AI chatbot interactions were statistically indistinguishable from human therapist conversations, with classifiers performing no better than chance. This suggests that users express emotional content toward AI in ways that mirror therapeutic norms typically seen in human-to-human care. These findings challenge prevalent assumptions about emotional superficiality in AI-based mental health tools, pointing instead to a pattern of emotionally coherent engagement that warrants serious consideration within broader mental health service delivery. While this evidence does not imply parity in therapeutic effectiveness, crisis management ability, or clinical expertise, it establishes foundational support for integrating AI chatbots as accessible adjuncts to human-delivered care within structured mental health pathways. The future of mental health support may increasingly involve hybrid models of digital and human care, with AI systems providing emotional scaffolding and preliminary support prior to or alongside clinician-based intervention.

Implications for Policy and Practice

The findings suggest that AI chatbots should not be positioned as replacements for licensed therapists but rather as supplementary tools that expand access to care, especially for individuals who face financial, geographical, or stigma-related barriers. Integration frameworks could include AI-based emotional triage, support during wait-list periods, reinforcement between therapy sessions, and guided self-reflection. Recent randomized control trials (e.g., MacNeill et al., 2024) demonstrate that clinically structured chatbots can produce measurable symptom relief, indicating their potential value as accessible first-line supports. However, policy must ensure these systems are implemented with clear disclosure protocols, risk boundaries, and escalation pathways. In line with APA (2024) recommendations, all mental health chatbots should clearly identify themselves as AI systems, state their limitations, apply protective data practices, include crisis referral guidance, and provide evidence of therapeutic grounding. To prevent therapeutic misconception, standardized regulatory frameworks similar to FDA guidelines for health technologies are needed to classify AI tools according to their clinical purpose, level of risk, and evidence base. AI mental health systems must be developed collaboratively by AI researchers, clinicians, ethicists, and patient representatives to ensure that emotional viability is paired with clinical responsibility. This study, therefore, informs both digital mental health innovation and policy reform, offering evidence-based justification for hybrid care models that integrate AI contributions while preserving the central role of human clinicians.

Limitations and Future Directions

Although this study provides rigorous and novel evidence, several constraints must be acknowledged. First, only text-based conversations were analyzed, excluding non-verbal cues, tone, and prosody elements known to strongly influence therapeutic effectiveness. Second, the study focused on emotional expression, rather than the quality or appropriateness of AI responses, meaning that clinical reasoning and therapeutic competence were not evaluated. Third, the datasets were limited to English-language and predominantly Western contexts, which restricts their transferability to culturally diverse populations. Fourth, the study employed a cross-sectional design, which prevented the observation of how therapeutic relationships may develop positively or negatively over time. Based on these limitations, future research should investigate longitudinal AI–user relationships, crisis response capacity, cultural variation in emotional communication, and direct comparison of response quality across AI systems and human therapists. There is also a pressing need for research on hybrid care pathways that combine AI support with therapist-based care, as well as structured guidelines for how clinical supervision and escalation protocols can be operationalised in AI-enabled mental health services. Ultimately, while emotional likeness is a necessary precondition for therapeutic applicability, future work must determine whether emotional engagement with AI genuinely supports recovery or merely resembles it.

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Data Availability All datasets analyzed during this study are publicly available through HuggingFace repositories. Analysis code and complete documentation are available upon reasonable request to the corresponding author. Processed de-identified data and analysis scripts will be made publicly available through the Open Science Framework (OSF) upon publication.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing interests The authors declare that the research was conducted without any commercial or financial relationships that could be construed as a potential conflict of interest.

Clinical Trial Number Not applicable.

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References

- Abd-Alrazaq, A., Safi, Z., Alajlani, M., Warren, J., Househ, M., & Denecke, K. (2024). AI chatbots for mental health: A scoping review of effectiveness, feasibility, and applications. *Applied Sciences*, 14(13), Article 5889. <https://doi.org/10.3390/app14135889>
- Amod (2023). Mental health counseling conversations [Dataset]. Hugging Face. https://huggingface.co/datasets/Amod/mental_health_counseling_conversations
- American Psychological Association (2024). Using generic AI chatbots for mental health support: A dangerous trend. APA Services. Retrieved March 12, 2025, from <https://www.apaservices.org/practice/business/technology/artificial-intelligence-chatbots-therapists>
- Arias, D., Saxena, S., & Verguet, S. (2022). Quantifying the global burden of mental disorders and their economic value. *eClinicalMedicine*, 54, 101675. <https://doi.org/10.1016/j.eclinm.2022.101675>
- Balcombe, L. (2023). AI Chatbots in Digital Mental Health. *Informatics*, 10(4), 82. <https://doi.org/10.3390/informatics10040082>
- Cross, S. P., Hermens, D. F., Scott, J., Salvador-Carulla, L., & Hickie, I. B. (2024). Opportunities and challenges in using artificial intelligence for mental health research and practice. *Australian & New Zealand Journal of Psychiatry*, 58(3), 189–194. <https://doi.org/10.1177/00048674231225591>
- Ettman, C. K., & Galea, S. (2023). Depression in the United States through the COVID-19 pandemic. *JAMA Network Open*, 6(9), e2332360. <https://doi.org/10.1001/jamanetworkopen.2023.32360>
- Grand View Research (2024). AI in mental health market size, share & trends analysis report. <https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-mental-health-market-report>
- Haque, M. R., & Rubya, S. (2023). An overview of chatbot-based mobile mental health apps: Insights from app description and user reviews. *JMIR mHealth and uHealth*, 11, Article e44838. <https://doi.org/10.2196/44838>
- Jacobson, N. C., Heinz, M. V., Mackin, D. M., Trudeau, B., Bhattacharya, S., Wang, Y., Banta, H., Jewett, A., Salzhauer, A., & Griffin, T. (2025). Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI*. <https://doi.org/10.1056/AIoa2400802>
- Khawaja, Z., & Bélisle-Pipon, J. C. (2023). Your robot therapist is not your therapist: Understanding the role of AI-powered mental health chatbots. *Frontiers in Digital Health*, 5, Article 1278186. <https://doi.org/10.3389/fdgth.2023.1278186>
- MacNeill, A. L., Doucet, S., & Luke, A. (2024). Effectiveness of a mental health chatbot for people with chronic diseases: Randomized controlled trial. *JMIR Formative Research*, 8, Article e50025. <https://doi.org/10.2196/50025>
- Moore, J., Haber, N., & Yamins, D. (2024). Exploring the dangers of AI in mental health care. *Stanford Human-Centered Artificial*

- Intelligence. Retrieved from <https://hai.stanford.edu/news/exploring-the-dangers-of-ai-in-mental-health-care>
- nbertagnolli (2023). Counsel Chat [Dataset]. Hugging Face. <https://huggingface.co/datasets/nbertagnolli/counsel-chat>
- Thompson, A., & Davies, C. (2024). It happened to be the perfect thing”: Experiences of generative AI chatbots for mental health. *npj Mental Health Research*, 3, Article 48. <https://doi.org/10.1038/s44184-024-00097-4>
- van Dijk, M. T., Murphy, E., Storfer-Isser, A., Rosen, M., Bruns, E., & Horwitz, S. M. (2023). Availability of mental health services for youth in the United States: 2017–2019. *Psychiatric Services*, 74(3), 236–243. <https://doi.org/10.1176/appi.ps.202200191>
- Williams, R., & Kumar, S. (2024). Human-human vs human-AI therapy: An empirical study. *International Journal of Human-Computer Interaction*, 40(12), 6841–6852. <https://doi.org/10.1080/10447318.2024.2385001>

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